MUAN MEURER

Pittsburgh, PA • muanmeurer@gmail.com • (626) 802-0269 • muanmeurer.github.io • linkedin.com/in/muanmeurer/

EDUCATION

Carnegie Mellon University - Pittsburgh, PA **December 2026 (expected)**
Master of Science in Mechanical Engineering | GPA 3.9/4.0
Relevant Coursework: *Electromechanical Systems Design, Mechatronic Design, Modern Control Theory, Robot Dynamics*

California State University, Los Angeles – Los Angeles, CA **May 2022**
Bachelor of Science in Physics | The Early Entrance Program | GPA 3.9/4.0

ENGINEERING PROJECTS

Lunar Robotics - NASA Competition, CMU | *Project Lead & Mechanical Engineer* **September 2025 - May 2026**

- Designed and integrated an autonomous excavation rover with a **73-person** multidisciplinary team and **$100,000 budget**, coordinating mechanical, electrical, and software interfaces through validation and competition.
- Selected mobility and excavation actuators and tuned PID controllers; maintained the integrated rover CAD assembly and coordinated mechanical interfaces across subsystems.
- Designed rover chassis and excavation structures in SolidWorks and evaluated them in ANSYS for an 80 kg payload and **1,500 N digging loads**, achieving a minimum calculated **factor of safety of 3** within an 80 kg rover mass limit.
- Authored system requirements, interface control documents, and risk and verification plans for SRR, PDR, and CDR; the team earned NASA's *Best 1st Year Team Presentation* award.

Automated Coil Winder - Westinghouse & CMU | *Student Mechatronics Engineer* **January 2026 - May 2026**

- Designed and built an automated variable-diameter helical coil winder for LVDT sensors in Westinghouse's eVinci microreactor, achieving measured **inductance within ±3% of target** across 40 winding configurations.
- Wired and integrated NEMA 17 stepper motors, motor drivers, and optical limit switches; programmed synchronized spindle rotation and linear traversal in Arduino C++ using micros()-based timing.
- Diagnosed initial winding-pitch variation caused by spindle–traverse timing mismatch and wire-nozzle misalignment; corrected synchronization and nozzle alignment to improve winding consistency.
- Developed a custom HTML user interface and integrated passive tensioning and interchangeable mandrels to produce 6-75 mm diameter coils using 18-32 AWG wire.

MoonRanger Lunar Mission - CMU | *Lead Mechanical Engineer* **June 2026 - Present**
Principal Investigator: Prof. William "Red" Whittaker

- Developed fabrication-ready CAD and engineering drawings for three rover and deployment-test assemblies; revised GD&T in rover-lander interface documentation and coordinated supplier quotes and procurement for deployment testing with Firefly Aerospace.
- Designed rover deployment-test hardware incorporating aluminum honeycomb, carbon fiber, bonded joints, and potted inserts; authored the test plan to verify rover-lander hold-down and release interfaces.
- Performed structural FEA for Falcon 9 launch loads and modal analysis in ANSYS, verifying the wheel assembly's first natural frequency at **139 Hz** against a **100 Hz** minimum requirement.

PROFESSIONAL EXPERIENCE

ABM & CMU | CSL Fellow – Digital Twins & Spatial Data **September 2026 - Present**

- Prototyped a Python computer-vision pipeline to identify facility assets in rendered 3D Gaussian-splat scans using Grounding DINO object detection and SAM2 segmentation.
- Combined multi-view images and camera-pose metadata to define spatial association and consolidate detection.
- Built a 3D point cloud asset-review prototype for inspecting and accepting, rejecting, or merging detections; evaluated IFC-based representation for downstream facility-management use.

SKILLS

Mechanical & Manufacturing: SolidWorks, CATIA, engineering drawings, GD&T (ASME Y14.5), CNC routing
Programming & Controls: Python, C++, MATLAB, Simulink, ROS2, Arduino, PID motor control
Analysis & Verification: ANSYS Workbench, structural and modal FEA, test planning